POTENTIOMETER-CONTROLLED LED BRIGHTNESS

SIMULATION LINK: https://wokwi.com/projects/478953520591089665

Objective

The objective of this project is to control the brightness of an LED using a potentiometer connected to an Arduino Uno. Instead of simply switching the LED ON and OFF, I designed the circuit so that the brightness gradually increases or decreases as I rotate the potentiometer.

Components

1. Arduino Uno – 1
2. Potentiometer – 1
3. LED – 1
4. 150-ohm resistor – 1
5. Breadboard – 1
6. Jumper wires – 5

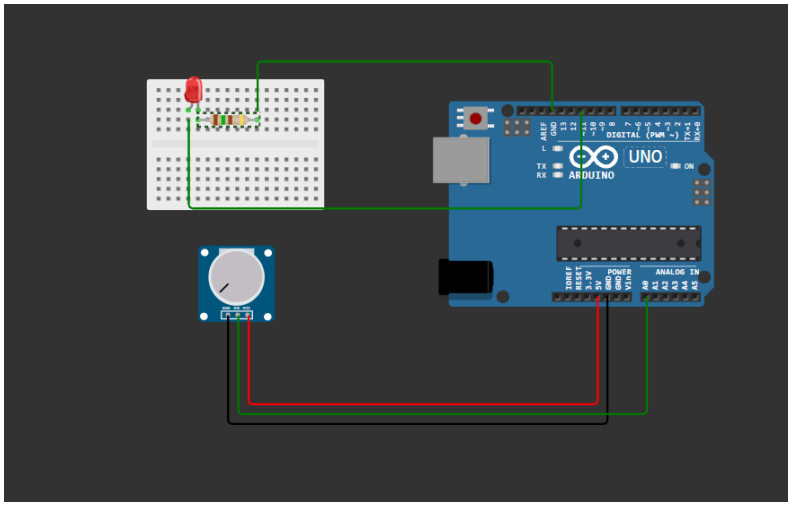
Circuit Connections

I connected the potentiometer to the Arduino Uno to read its varying resistance as an analog input.

- One outer terminal of the potentiometer is connected to 5V.
- The other outer terminal is connected to GND.
- The middle terminal of the potentiometer is connected to analog pin A0 of the Arduino.
- The positive (longer) leg of the LED is connected to digital pin 11.
- The negative (shorter) leg of the LED is connected to GND through a 150-ohm resistor..

I used digital pin 11 because it supports pulse width modulation, which allows me to control the brightness of the LED using the potentiometer.

Circuit Diagram



Arduino Code

I wrote the following Arduino program to read the potentiometer value and control the LED brightness.

```
int potPin = A0;

int ledPin = 11;

int potValue = 0;

int brightness = 0;

void setup() {

  pinMode(ledPin, OUTPUT);

}

void loop() {

  potValue = analogRead(potPin);

  brightness = map(potValue, 0, 1023, 0, 255);

  analogWrite(ledPin, brightness);

  delay(1);
```

```
}
```

Working Principle

When I rotate the potentiometer, its output voltage changes. The Arduino reads this voltage through analog pin A0 using the `analogRead()` function.

The analog reading ranges from 0 to 1023. A reading close to 0 represents a low input voltage, while a reading close to 1023 represents a voltage close to 5V.I used the `map()` function to convert this range into a value between 0 and 255, which is suitable for controlling the LED using PWM.

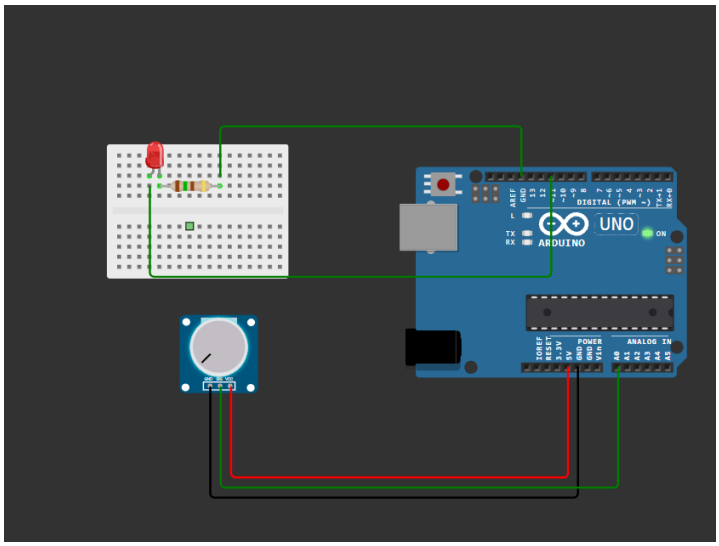
The `analogWrite()` function generates a PWM signal on pin 11. By varying the PWM duty cycle, I can control the power delivered to the LED and thereby change its brightness.When the potentiometer is rotated towards the lower reading, the LED becomes dimmer.

When it is rotated towards the higher reading, the LED becomes brighter.The `delay(1)` function introduces a short delay of 1 milliseconds between consecutive readings, allowing the program to update the brightness continuously.

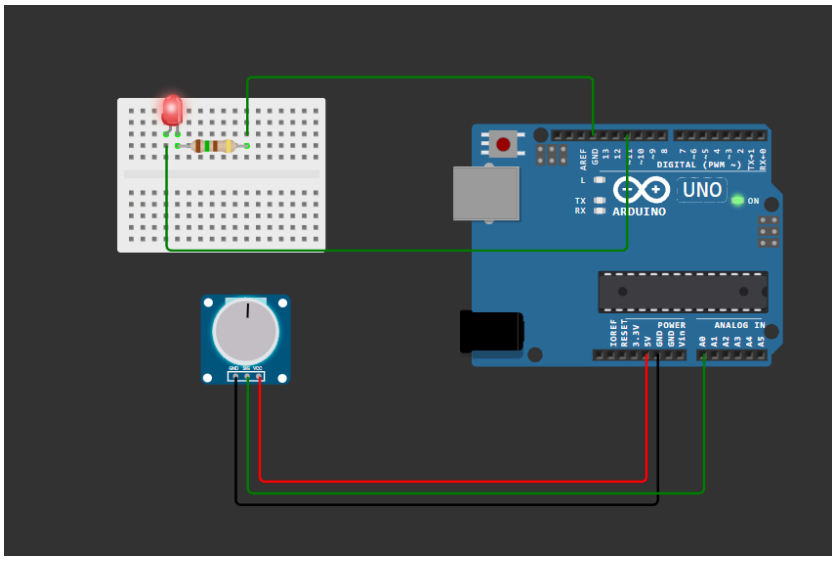
Output

The LED brightness changes gradually rather than simply switching between fully ON and fully OFF.

Output Screenshot 1: Off



Output Screenshot 2: Medium Brightness



Output Screenshot 3: On /High Brightness

